COMPILE TIME

Is the syntax correct, all variables being referenced properly declared, etc?

- No → compiler error
- Yes →

Is there a cast on this line?

- Yes →

We have a var w static type ___S___ being cast to the class ___C___. Is a C an S, or is an S a C? In other words, are C & S on some inheritance path?

- No → compiler error

Yes. The cast was successful. For the remainder of this line, we'll say this var has static type C.

- No →

Is there a method call?

- Yes →

We have a variable of static type ___S___ calling a method named ___m___ taking in parameter(s) of type ___p___. Does the class S have a method m that takes parameters p?

- Yes → Cool, let's jot down that method signature. We'll look at this again at runtime.
- No → Does a parent class of S have a method m that takes parameters p?
  - Yes → Cool, let's jot down that method signature. We'll look at this again at runtime.
  - No → Does S have a method m that takes in a parameter of type q, where q is a superclass of p? In other words, S has a method m(q) that would accept a p because a p is a q?
    - Yes → Cool, let's jot down that method signature. We'll look at this again at runtime.
    - No → Does a parent class of S have a method m that takes in type q where q is a superclass of p?
      - Yes → Cool, let's jot down that method signature. We'll look at this again at runtime.
      - No → compiler error

- No →

Is there an assignment?

- Yes →

We are assigning a variable of static type ___S___ to some value on the right side with static type ___V___. Is a V an S? In other words, does V inherit from S?

- Yes → Cool, the line compiles! Now, onto runtime!
- No → compiler error

- No → Cool, the line compiles! Now, onto runtime!

RUNTIME

Was there a cast on this line?

Yes

We casted a variable of dynamic type ____ (D) to the class ____ (C).

Is a D a C? In other words, does D inherit from C?

No

Runtime error :C
Lying cast

Yes

No

Is there a method call on this line?

Yes

We have a variable V of dynamic type ____ (D) calling a method. Recall that during compile time, we jotted down a method signature, we'll call it m. Does the class D have a method with the <u>exact</u> <u>same</u> <u>signature</u> as m, including the parameter types?

No

Go up the inheritance tree until you find one that does. Execute the method found.

Yes

Execute that method in D.

No

Yay! Hope there are no other runtime errors (like divide by zero, for example)